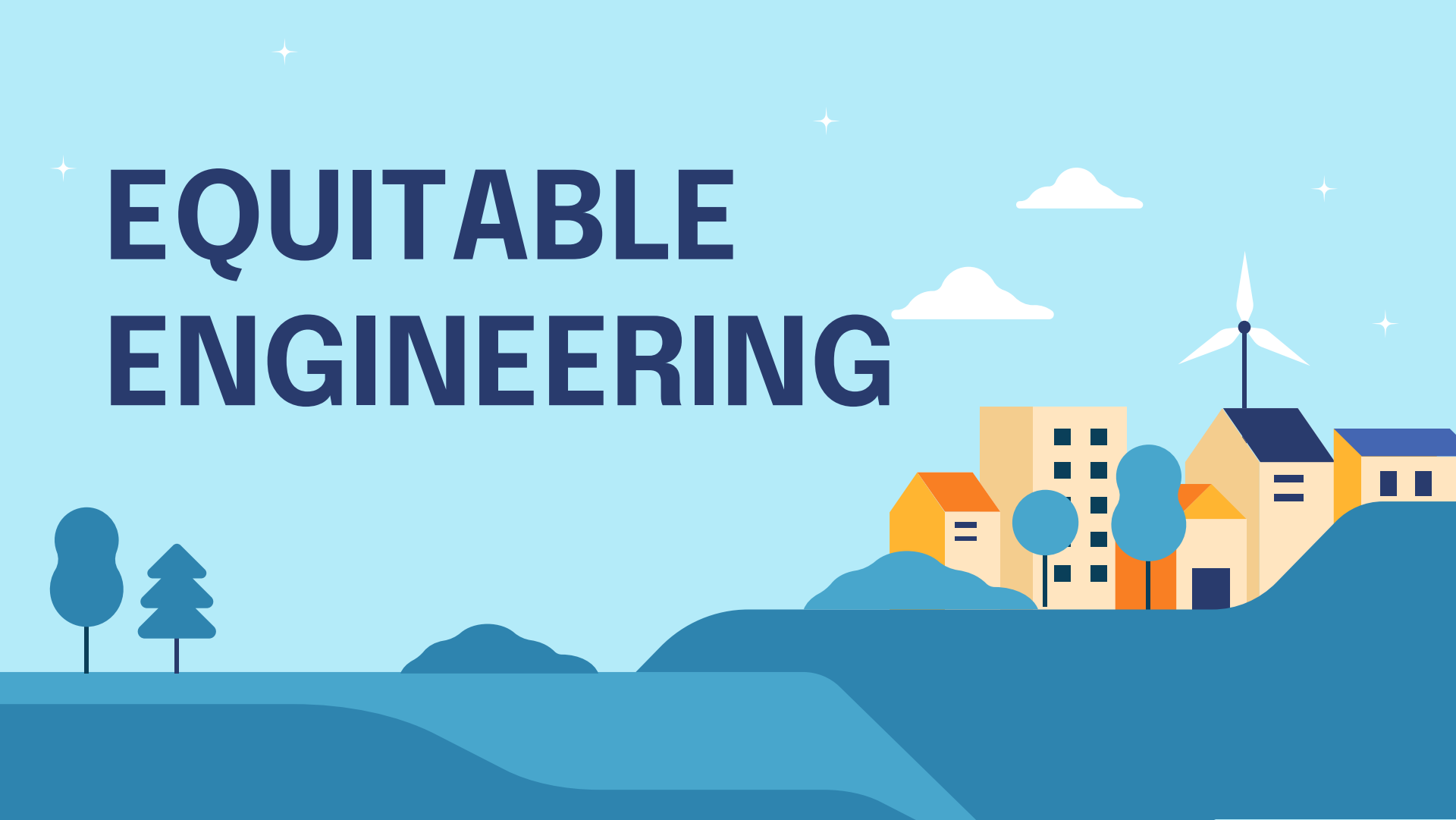
EQUITABLE ENGINEERING IN CROSS-CULTURAL SETTINGS

Maya Struzak, B.S. Civil Engineering '24

EQUITABLE ENGINEERING



FACETS

- Minimize Environmental Impact
- Engage Communities
- Promote Social Equity
- Address Ethical Considerations

EQUITABLE ENGINEERING

INTER-DISCIPLINARY

CREATIVITY

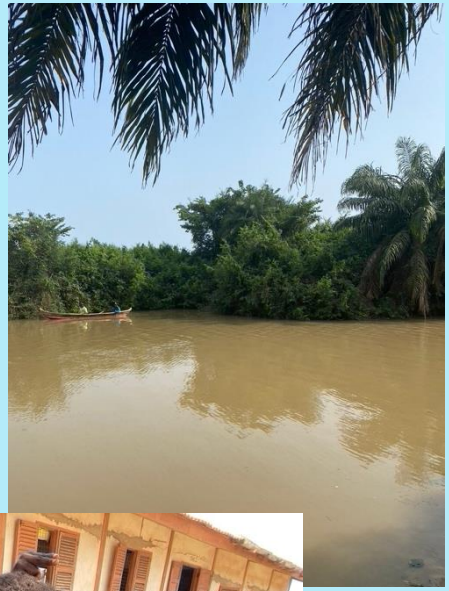
SOCIAL CONCSIOUSNESS

MULTI-CULTURAL

INNOVATION

Portland, OR
USA
Anomabo,
Ghana

MS



SOCIAL CONSCIOUSNESS

Key Takeaways:

01

Impact

Engineering has a direct impact on a community's health and QOL

02

Involvement

Community stakeholders should be involved in the design process

03

Context

Designs should be context specific and culturally sensitive

Reykjavik,
Iceland
Portland, OR
USA

Social Business Model Canvas

Social Innovation Lab

Key Resources

Physical: location, utilities, plant
Human: construction, maintenance, operation
Intellectual: informational webpage, internal database
Finance: private purchase agreement, government loan (Justice40)

Partners + Key Stakeholders

- Government (BOEM, DOE)
- Community members
- Competing utility providers
- Environmental protection agencies

Key Activities

- Electricity production
- Updating of production techniques
- Platform management
- Community involvement
- Employment

Type of Intervention

Public electricity service

Channels

- Direct communication with utility companies and government
- Public engagement

Segments

- Utilities
- Customers

Value Proposition

- Reduction in emissions
- Energy security
- Subsidized energy to increase affordability (Justice40)

Cost Structure

Fixed

- Labor
- Materials
- Licenses
- Easements

Variable

- Operation
- Maintenance

Surplus

- Maintenance
- Technology innovation (R&D)

Revenue

- Electricity sales to utility company
- Tax credits
- Federal grants and incentives

Inspired by The Business Model Canvas

INNOVATION

Key Takeaways:

01

Implementation

Important to consider economic, social, and practical aspects

02

Cost

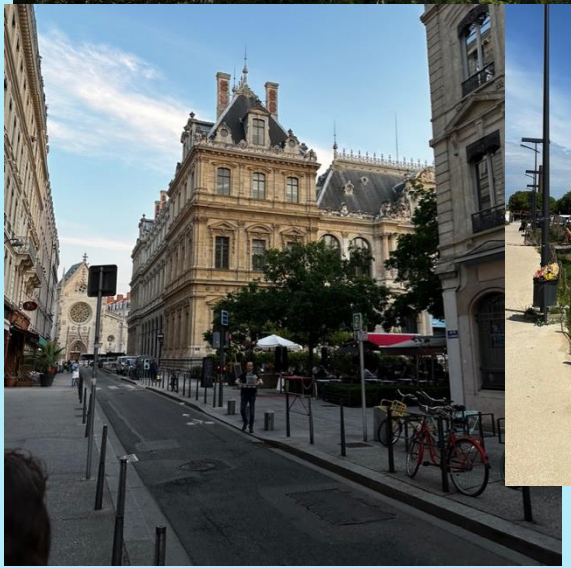
Understanding cost and project lifestyle are essential

03

Sustainability

Goal should be to achieve both economic and environmental sustainability

Portland, OR
USA
Lyon,
France



MULTICULTURAL

Key Takeaways:

01

Perspectives

Important to consider various social, environmental, and cultural perspectives in design

02

Learning

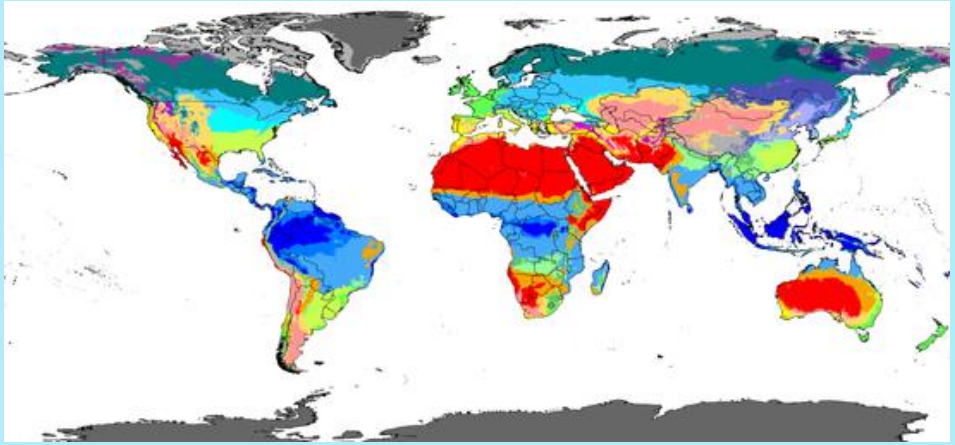
There is so much to learn from people different than ourselves

03

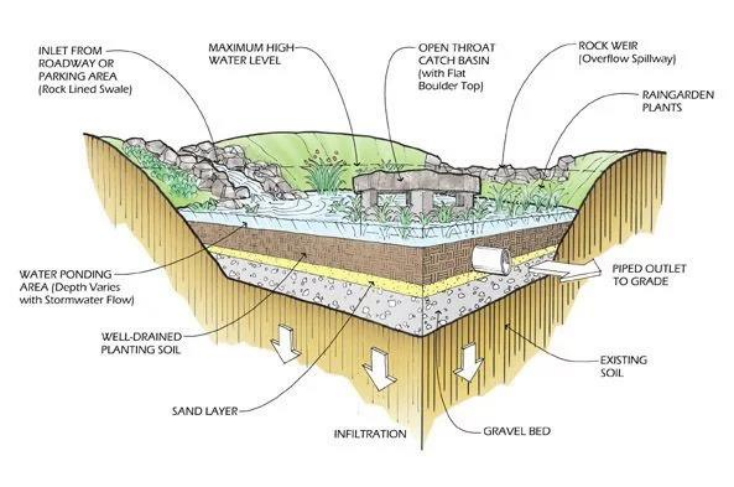
Collectivity

Tackling infrastructure challenges takes a collective effort

via Thornton Tomasetti



via Wikipedia.org



via Bluegrass Lawn Care

via Learn Hydrology Studio

CREATIVITY

Key Takeaways:

01

Representation

Lack of studies outside of USA, Europe, and China

02

Context

Functioning of bioretention systems vary by geographic location

03

Adaptability

Engineers must account for unprecedented changes in climate and development

Portland, OR
USA

INTERDISCIPLINARY: ETHICS

Deontological

Following duty

Utilitarianism

Weighing the costs and benefits

Virtue

Applying personal standards

Care

Practicing compassion

CONCLUSION

01

Versatility

How to be a well-rounded engineer

02

Impact

Develop solutions to better the world around me

03

Justice

Address global issues through an equity lens

THANKS!

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